

Optimizing ResNet-50 for Multi-Class Diabetic Retinopathy Grading: An Explainable AI Approach

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Abstract— Diabetic Retinopathy (DR) is a leading cause of vision loss among the working-age population, driven by the rising global prevalence of Diabetes Mellitus, which affected over 529 million people in 2021. With a critical shortage of ophthalmologists in many regions, automated screening solutions are essential. This study presents a deep learning framework utilizing a fine-tuned ResNet-50 architecture to classify DR into five severity stages using an APTOS dataset. To address severe class imbalance, we implemented a robust preprocessing pipeline involving circular cropping, Contrast Limited Adaptive Histogram Equalization (CLAHE), and synthetic data augmentation to balance the training set at 2,000 images per class. The model was optimized using a two-phase transfer learning schedule, incorporating L2 regularization and dropout to mitigate overfitting. Rigorous 5-fold cross-validation demonstrated robust and highly significant performance (all $p < 0.001$) compared to established baselines, achieving a mean accuracy of 84.51%, QWK of 0.8226, F1-score of 84.43%, and ROC AUC of 0.9749. Furthermore, specificity remained consistently high across all severity stages (0.94–0.99), indicating reliable differentiation of subtle disease stages with minimal false positives.

Keywords—Diabetic Retinopathy, Deep Learning, ResNet-50, Transfer Learning, Data Augmentation, Explainable AI, Medical Image Classification

I. INTRODUCTION

Diabetes Mellitus (DM) is a chronic medical condition wherein the body is unable to produce enough insulin or cannot use it properly, resulting in high blood sugar levels. More than 529 million people were affected by DM worldwide in 2021, and this number is projected to increase to 1.31 billion by 2050, making it a continuous global burden [1]. DM has numerous complications in the human body, with one of them being Diabetic Retinopathy (DR). DR is a microvascular complication of DM and a leading cause of vision loss among working-age populations caused by high blood sugar levels [2][3]. One well known process of this damage occurs by the apoptosis of pericytes, which stabilize blood vessels and regulate blood flow. Since these cells are responsible for maintaining structural support for capillaries, their loss could lead to localized outpouching of these walls, which is associated with

microaneurysm (MA) formation, a key characteristic of DR [3]. Other key lesions in diabetic retinopathy (DR) include: Hemorrhages (HR), which appear as dark red or black blotches caused by ruptured blood vessels; Hard Exudates (HE), irregularly shaped yellow lesions formed when lipoprotein and macrophage complexes leak from capillaries; Soft Exudates (SE), or "cotton-wool spots," which are areas of ischemia resulting from blocked retinal arteries; and Neovascularization (NV), the formation of new blood vessels in response to extreme hypoxic conditions [4], [5], [6], [7]. Symptoms of DR include lots of vision-related symptoms such as dark spots or strings, blurred vision, fluctuating vision, empty spots, vision loss, and difficulty with color perception [8]. Diabetic Retinopathy can be classified into Non-Proliferative Diabetic Retinopathy (NPDR) and Proliferative Diabetic Retinopathy (PDR) based on severity. According to the International Clinical Diabetic Retinopathy (ICDR) scale, classification is split into No DR, Mild NPDR, Moderate NPDR, Severe NPDR, and PDR [9]. In Mild NPDR, there are only MAs present. The clinical definition of Moderate NPDR includes microaneurysms accompanied by retinal hemorrhages, hard exudates, or cotton wool spots. In this classification, there should be no signs of severe NPDR. Severe NPDR has more requirements: more than twenty intraretinal hemorrhages in all 4 quadrants, definite venous beading in 2 or more quadrants, prominent intraretinal microvascular abnormality in 1 or more quadrants, and no sign of PDR. Finally, in PDR, there must be new vessel formation and vitreous/pre retinal hemorrhages, causing bleeding and fluid to build up in the back of the eye.

The burden of DR in certain parts of the world is immense, and the issue is extenuated by the lack of specialists to detect eye disorders [10]. For instance, in India there is 1 diabetic retinopathy specialist per 1.26 million patients. This demonstrates a need for more accessible diagnosing procedures [11]. Deep Learning (DL) is a rapidly advancing subfield of Machine Learning (ML) that leverages neural network architectures, such as Convolutional Neural Networks (CNNs), to automatically learn and identify complex patterns in large datasets [12]. Specifically, these models have been applied to various medical scans to identify patterns between images and reduce fatigue of human experts, ultimately improving disease detection [12]. CNN models like ResNets, InceptionNets, VGGs, and EfficientNets have all been used in the classification of stages of DR [13], [14], [16]. In this paper,

a pre-trained ResNet-50 model on the ImageNet dataset is used and fine-tuned to provide optimal performance on the Asia Pacific Tele-Ophthalmology Society 2019 Blindness Detection (APTOS 2019 BD) dataset. ResNet-50 is a model that utilizes residual connections to solve the vanishing gradient problem that is observed with extremely deep neural networks [25].

II. RELATED WORKS

A. Custom-Built Architectures

A model called RSG-Net was developed for both binary detection of DR and multi-class classification depending on user input. Using the Messidor-1 dataset, image preprocessing techniques like cropping, denoising, Histogram Equalization (HE), and resizing were done on each image. Later, data augmentation techniques like flips, zooms, rotations, contrast adjustment, color adjustment, and brightness adjustments were utilized. Upon developing the CNN model with ReLU activation functions to extract edges, textures, and patterns from images, validation metrics for both binary and multi-class classification were analyzed. RSG-Net achieved a test accuracy of 99.36% for 4-stage classification and 99.37% for binary classification, outperforming most models. However, the model has possible risk of overfitting because of high validation metrics observed; hence, further work is needed using regularization techniques to reduce possible overfitting [2].

B. Ensemble Learning

Ensemble learning refers to the use of multiple models to produce a single, more accurate predictive model. An ensemble DL and EfficientNet methodology is used for the multi-class classification of DR. By classifying the severity of diabetic retinopathy using the EfficientNetB0 model, the study utilized a pre-trained version of the model, modifying it by replacing the top layer with a new layer designed to classify retinal images into five distinct severity categories.

The framework was developed using a large, imbalanced Kaggle dataset of over 35,000 images. To create a more effective training environment, the researchers balanced the dataset through under sampling, resulting in a smaller, more evenly distributed set of 3,704 images. Preprocessing steps included resizing all images to a uniform 224x224 dimension, normalizing pixel values, and applying data augmentation techniques such as random flips, rotations, and zooms to prevent overfitting [12].

C. Transfer Learning

Transfer learning involves repurposing a model trained on a large, general dataset for a new, specialized task. A study by Gulshan et al. utilized a well-known Inception-v3 architecture for multi-class classification. It was pre-trained on the vast ImageNet dataset containing millions of everyday images. This pre-trained model was then fine-tuned on a massive development dataset of 128,175 retinal fundus photographs. The resulting algorithm was validated on two independent clinical datasets (EyePACS-1 and Messidor-2), achieving an AUC of 0.99. At an operating point optimized for screening, the model reached a high sensitivity of 97.5% and a high specificity of 93.4% [13]. Another research study focused on improving the effectiveness of ResNet-50 on DR severity classification highlights the necessity for L1/L2 regularization and dropout to

prevent overfitting [17]. Furthermore, by using Grad-CAM, Huang et al. was able to visualize the results of the model to understand its decision making.

III. METHODOLOGY

A. Dataset and Problem

The publicly available Diabetic Retinopathy 224x224 (2019) dataset hosted on Kaggle (Sovit Rath) was used [18]. It consists of 3662 color fundus images at 224×224 resolution, which were labelled into five clinical grades: 0: No_DR, 1: Mild, 2: Moderate, 3: Severe, 4: Proliferative_DR. The class distribution was 1805 No_DR (49.3%), 370 Mild (10.1%), 999 Moderate (27.3%), 193 Severe (5.3%), and 295 Proliferative_DR (8.1%). The set was strongly imbalanced, and the largest class was about 9.4 times the smallest. Such an imbalance increases the likelihood of strong bias in the model predictions, with the favor going towards the No_DR dataset because it comprises most of the dataset.

B. Preprocessing Pipeline

Each image underwent four sequential operations. First, true circular cropping was carried out by converting the image to greyscale, applying a 5×5 Gaussian blur, and thresholding at 10 to obtain external contours. The largest contour was taken as the fundus; its minimum enclosing circle was computed; a filled circular mask was created and applied bitwise to remove non-retinal pixels. Finally, the retina was cropped to the bounding rectangle. This removed background artifacts such as black borders, restricted the field to retinal tissue, and improved consistency among all images. Second, the cropped image was resized to 224×224 using area-based interpolation (INTER_AREA). Cropping changed spatial dimensions, so resizing standardized the tensor shape required by ResNet-50 and stabilized batching during training. A popular preprocessing technique is Histogram Equalization (HE), which takes the most frequently occurring intensity values in the histogram and spreads it out, improving the contrast among parts of the image but also over-amplifying noise. To mitigate the over amplification issue, Contrast-Limited Adaptive Histogram Equalization (CLAHE) was used by clipping the histogram of intensities at a specific value [19].

C. Data Splits and Data Loading

A stratified split reserved 20% of images for validation while preserving the marginal label distribution. The split was fixed with `random_state = 42` to support reproducibility. Images were presented to the network at 256×256 pixels. Although the preprocessed corpus was stored at 224×224 , inputs were resampled to 256×256 at load time. This choice increased the final spatial grid in ResNet-50 from 7×7 to 8×8 , providing slightly finer spatial support for lesion patterns, and reduced aliasing under geometric augmentations such as $\pm 25^\circ$ rotations. No new detail was created by this upsampling; the resize standardized the working resolution for training and was compatible with memory limits through gradient accumulation. For the training partition, we apply the data augmentation sequence mentioned in the below section. The validation partition employed only Resize (256×256) followed by the same

normalization. Mini-batches of size 32 were drawn with shuffling enabled for training and disabled for validation. Unless otherwise specified, PyTorch’s default DataLoader worker and pinned-memory settings were used.

D. Data Augmentation and Training Objectives

Due to the severe class imbalance, we adopted two complementary strategies: dataset-level balancing by augmentation and class-weighted optimization. Dataset-level balancing by augmentation was done after the test/train split. We have constructed a balanced set with a target of 2,000 images per class. For each class with fewer than 2,000 images, we synthesized additional samples by applying label-preserving image transformations to preprocessed images until the target was reached. Transformations were stochastic and drawn from a family comprising horizontal reflections, small in-plane rotations of approximately $\pm 11^\circ$, slight zoom-out (up to 10%), and contrast adjustments. These operations preserved lesion morphology while increasing appearance diversity and viewpoint variation. This procedure generated 7047 synthetic images in total: 904 for No_DR, 1630 for Mild, 1,001 for Moderate, 1,807 for Severe, and 1,705 for Proliferative_DR, yielding 2,000 instances per class and 10,000 training images overall. Through balancing the size of classes, we reduced the risk of biased decision thresholds and improved calibration for minority classes. The PyTorch validation loader did not apply any additional online augmentation beyond resizing and normalization. Training objective with class weighting. Let $z \in \mathbb{R}^5$ denote the network logits for an input and let $p = \text{softmax}(z)$. Class weights were computed from the training labels as $w_m = \frac{N}{(Knk)}$ where N is the number of training examples, $K=5$ is the number of classes, and nk is the count for class k . Let $D = \{(x_i, y_i), i = 1, \dots, N\}$ denote the training set, where x_i is the input image and y_i is the corresponding ground truth label. There are a variety of objective functions that can be used to measure the discrepancy between the predicted probability distribution $\hat{y}_i$ and the ground-truth distribution $\tilde{y}_i$ (one-hot encoded ground truth y_i). The cross-entropy loss is the most used loss function for classification tasks, which is the negative log-likelihood of a Bernoulli or categorical distribution. However, on imbalanced datasets, plain cross-entropy can be determined by majority classes, because each training example contributes equally to the loss and errors in rare classes receive relatively little weight. Therefore, a weighted cross-entropy loss function was used where the loss of an individual image was $L_i = - \sum_{j=1}^C w_j y_{(i,j)} \log(\hat{y}_{(i,j)})$. In this equation, C represents the number of classes (5 in our case), i is the sample index of a specific image in a batch, j is the index of the class, $y_{(i,j)}$ is the “one-hot” indicator, $\hat{y}_{(i,j)}$ is the predicted probability that image i belongs to class j , and w_j is the class weight. This weighted cross entropy function increases the penalty for errors on minority classes and reduces it for majority classes. The same weighted cross-entropy loss was used when optimizing on the training and validation splits, while evaluation metrics (accuracy, F1-score, QWK) were computed directly from predictions without class-weight corrections. Training-time augmentation. In addition to the dataset-level balancing above, the training loader applied similar online

transformations in every epoch. Inputs were resized to 256×256 , followed by stochastic horizontal and vertical flips with probability $p=0.5$, random in-plane rotation sampled uniformly from $[-25^\circ, 25^\circ]$, and photometric jitter comprising brightness 0.2, contrast 0.2, and saturation 0.1. Images were then converted to tensors and normalized channel-wise using the ImageNet statistics. The validation loader used only resizing and normalization to give accurate validation metrics. This separation ensured that performance estimates reflected generalization rather than exposure to augmentation noise.

E. Model

We employed a ResNet-50 convolutional neural network initialized with ImageNet-pretrained weights (torchvision ResNet50_Weights.DEFAULT). ResNet-50 is a deep residual convolutional architecture that organizes layers into residual blocks with identity or projection shortcuts to match input and output dimensionality. These shortcuts allow the blocks to learn residual functions, which improves optimization stability and enables substantially greater depth without vanishing-gradient failure where the network’s gradient is reduced upon adding more layers. Backbone configuration. The network comprised a stem with a 7×7 convolution (stride 2, 64 channels) followed by batch normalization, ReLU, and a 3×3 max-pool (stride 2).

F. Optimization and Schedules

Training phases and freezing. The optimization proceeded in two phases. In Phase 1 (feature extraction) all backbone parameters were frozen and only the classification head was trained. In Phase 2 (fine-tuning) all layers were unfrozen and optimized end-to-end. Optimizers and hyperparameters. Phase 1 used Adam on the head parameters with learning rate $\eta = 1 \times 10^{-3}$ ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$). Phase 2 used Adam on all parameters with $\eta = 1 \times 10^{-5}$ and weight decay $\lambda = 1 \times 10^{-4}$ (L2 regularization), with the same β_1 , β_2 , ϵ . No learning-rate scheduler was employed. Adam is an adaptive stochastic optimizer that maintains first- and second-moment estimates of the stochastic gradients. Let θ denote parameters and g_t the gradient at step t . Adam forms exponential moving averages $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$, $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t$, applies bias corrections to these estimates: $\hat{m}_t = \frac{m_t}{(1 - \beta_1^t)}$, $\hat{v}_t = \frac{v_t}{(1 - \beta_2^t)}$, and updates parameters as $\theta_{t+1} = \theta_t - \eta_{\theta} \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$. Here η is the learning rate; β_1 and β_2 (typically 0.9 and 0.999) set the timescales of the momentum-like first moment and the adaptive second moment; ϵ (typically 1×10^{-8}) prevents division by zero and improves numerical stability. In our Phase 2, a weight-decay penalty $\lambda = 1 \times 10^{-4}$ was used with PyTorch’s Adam, which corresponds to adding $\lambda \theta_i$ to the gradient (classical L2 regularization). Epochs and early stopping. Phase 1 ran for 10 epochs. Phase 2 ran for up to 50 epochs with early stopping (patience 10 epochs) based on validation loss. The best model (lowest validation loss) was checkpointed and restored for final evaluation. Gradient accumulation and effective batch size. Phase 2 used gradient accumulation with ACCUMULATION_STEPS = 4. With a physical batch size of 32, the effective batch size was 128. With the training images, there were 250 mini-batches per epoch in Phase 2; accumulation yielded 63 optimizer updates per epoch (i.e.

[250/4]), compared with 250 updates per epoch in Phase 1 (no accumulation).

G. Evaluation Protocol

Validation setting. All reported results were obtained on the stratified validation split (20% of the balanced corpus). The network was set to evaluation mode (Dropout disabled; batch normalization in inference mode). No test-time augmentation was applied; validation preprocessing comprised only of resizing and normalization (Section 2). Predictions and probabilities. For each input image x , the network produced logits $z \in \mathbb{R}^5$ and class probabilities $p = \text{softmax}(z)$. Predicted labels were defined as $\hat{y} = \text{argmax}_{c \in \{1, \dots, 5\}}$.

Metrics. Let $\{(x_i, y_i)\}_{i=1}^m$ denote a validation batch with $K = 5$ classes and let $1[\cdot]$ be the indicator. Top 1 accuracy. $\text{Acc} = \frac{1}{m} \sum_{i=1}^m 1[\hat{y}_i = y_i]$. Per-class precision/recall/F1. For class k : $\text{Prec}_k = \frac{TP_k}{TP_k + FP_k}$; $\text{Rec}_k = \frac{TP_k}{TP_k + FN_k}$; $F1_k = 2 \cdot \text{Prec}_k \cdot \text{Rec}_k / (\text{Prec}_k + \text{Rec}_k)$. We also report macro and weighted aggregates as provided by scikit-learn's classification report: macro averages treat classes uniformly (unweighted mean over k), whereas weighted averages use class supports s_k . A $K \times K$ confusion matrix C with entries C_{ij} counting samples of true class i predicted as j . We visualized C with a heatmap (counts); a row-normalized variant $\frac{C_{ij}}{\sum_j C_{ij}}$ can be added for error-profile analysis. With ordinal labels 0,1,2,3,4 QWK measures agreement beyond chance with a quadratic penalty for larger ordinal discrepancies. Let O be the observed $K \times K$ confusion matrix normalized to probabilities and let E be the expected matrix under label independence. With weights $w_{(i,j)} = \left(\frac{i-j}{K-1}\right)^2$, QWK is $k_w = 1 - \frac{\sum_{(i,j)} w_{(i,j)} O_{(i,j)}}{\sum_{(i,j)} w_{(i,j)} E_{(i,j)}}$. Values range from -1 (complete disagreement) to 1 (perfect agreement), with 0 indicating chance-level agreement. For each class k , we computed ROC _{k} from the binary problem $1[y = k]$ vs. probability p_k and its area AUC _{k} . We further computed the ROC and micro-AUC by aggregating decisions across all classes (label_binarize and ravel). A macro-average AUC can be obtained as the unweighted mean of $\{\text{AUC}_k\}$, although it was not explicitly printed by the evaluation script. Class-Wise Confidence The predicted probability for the correct class from the softmax function was plotted on the x-axis with the y-axis representing the density. The sharper the peaks towards the right, closer to 1.0 in the x-axis, the more the model is certain about its predictions Model selection and early stopping. Training tracked validation weighted cross-entropy loss each epoch. The best model was defined as the checkpoint with the lowest validation loss; early stopping with patience 10 halted optimization when no further improvement was observed. Final metrics were computed using this best checkpoint. Learning curves and recorded statistics. We recorded per-epoch training and validation losses as well as accuracy, precision, recall, F1 (weighted), and multiclass AUC (OvR, macro). Plots to the predicted class at the chosen layer; it is a qualitative tool and does not constitute lesion segmentation. Because saliency resolution is limited by the target layer's stride, it can be sensitive to background artifacts. We used circular cropping to reduce this noise, but we treated the Grad-CAM results

primarily as qualitative support for our classification metrics. We evaluated using 5-fold cross validation and statistical t testing.

H. Interpretability (Grad-CAM and SHAP)

We also generated localization maps using Grad-CAM. We denoted the activation map of channel k at the selected convolutional layer and formed the heatmap using ReLU functions. The heatmap was min-max normalized to $[0,1]$, resized to 256×256 , and visualized by linear blending with the RGB image. Heatmaps were rendered with a JET colormap and blended with opacity $\alpha = 0.35$. Layer and class selection. The target layer was the last bottleneck block in stage conv5_x (PyTorch: layer4[-1]), which preserved spatial detail while remaining semantically deep. By default, the target class c was the model's predicted label ($c = \text{argmax softmax}(z)$); a specific class could also be requested. Gradients were obtained by backpropagating a one-hot vector for class c through the network in evaluation mode. Grad-CAM highlights spatial regions that contributed positively to the predicted class at the chosen layer; it is a qualitative tool and does not constitute lesion segmentation. Because saliency resolution is limited by the target layer's stride, it can be sensitive to background artifacts. We used circular cropping to reduce this noise, but we treated the Grad-CAM results primarily as qualitative support for our classification metrics.

We also generated SHAP visualizations using SHAP values and assigning importance value to each feature for a single model prediction, showing how much it contributes to the final output. SHAP values are generated through combining various interpretability techniques such as LIME, DeepLIFT, and Layer-Wise Relevance Propagation. Specifically, in image analysis, the features are the pixels of the image and dictate both pixels which are positive and negative for model diagnosis allowing a more comprehensive understanding of the model's vision. The importance of SHAP values has been utilized throughout the medical field, especially in image analysis. For instance, SHAP has been used as an xAI technique extensively in histopathological images and progression of brain tumors.

IV. RESULTS

We evaluated the fine-tuned ResNet-50 on a set separated by disease grade. It was found to have an overall accuracy of 84.5% and a quadratic weighted kappa (QWK) of 0.82, indicating close agreement to reference grades on the five scale.

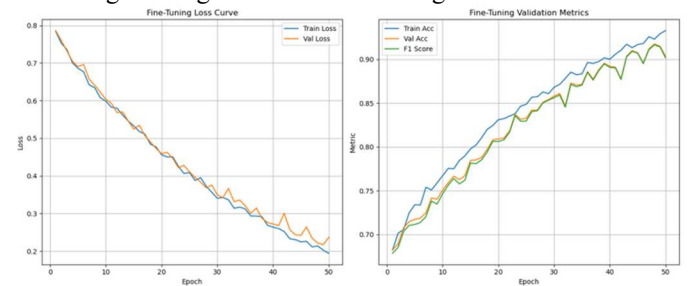


Figure 1 Fine Tuning Loss and Validation Metrics Over Epochs

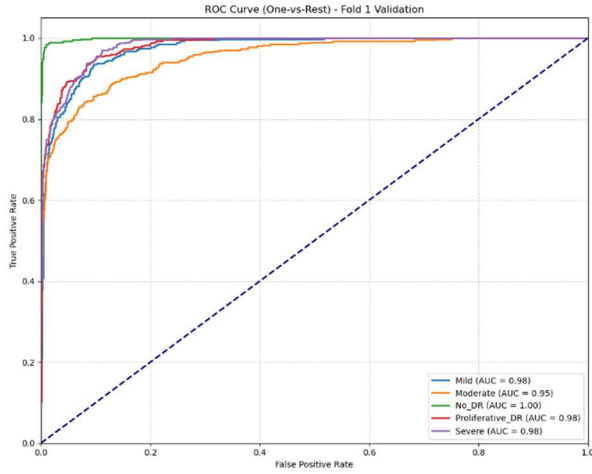


Figure 2 ROC Curves for Each Class (Fold 1)

In Figure 2, the high average ROC AUC of 0.98 suggests that the network is very careful in assigning positive labels and that most errors arise from subtle disagreements among closely ranked grades.

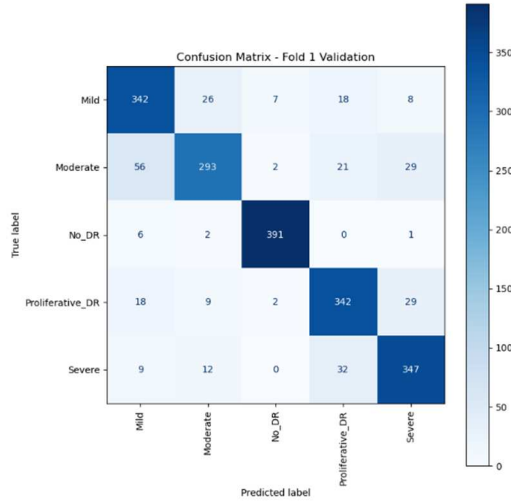


Figure 3 Confusion Matrix (Fold 1)

Specificity remained high across all classes. In Figure 3, the confusion matrix shows that the true negative rates were as follows: 0.94 for Mild, 0.97 for Moderate, 0.99 for No_DR, 0.96 for Proliferative DR, and 0.96 for Severe DR. This indicates that false positives occur infrequently, although there are some class-based differences. To evaluate generalizability, a 5-fold cross-validation was conducted, yielding stable results with low variance across all data splits. The model achieved a mean Accuracy of 0.8451 (± 0.0097), Weighted F1 of 0.8443 (± 0.0097), Precision of 0.8457 (± 0.0094), Recall of 0.8451 (± 0.0097), ROC AUC of 0.9749 (± 0.0024), and a Quadratic Weighted Kappa (QWK) of 0.8226 (± 0.0144). One sample t-tests confirmed that performance across all metrics was highly statistically significant compared to established baselines (all $p < 0.001$), demonstrating robust and consistent predictive capability.

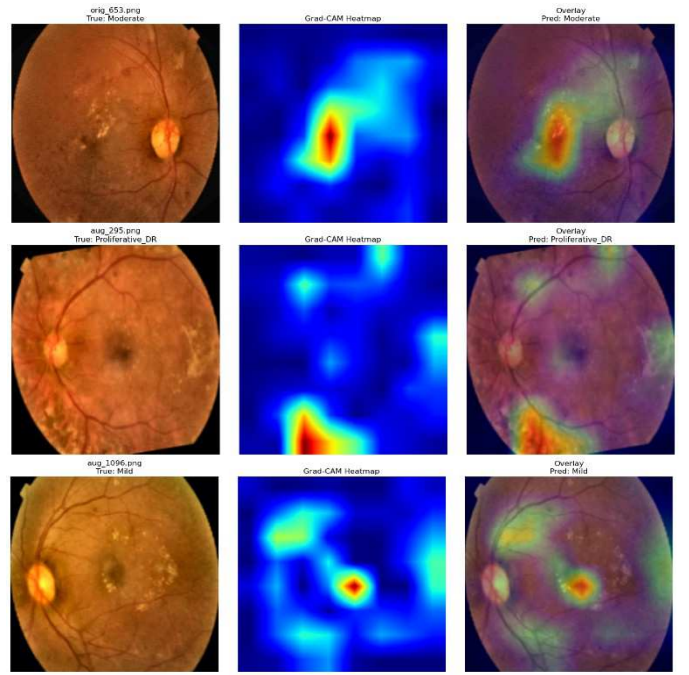


Figure 4 Grad-CAM Maps of Mild, Moderate, and Proliferative DR

A qualitative check of Grad-CAM heatmaps (Figure 4) revealed that for correctly classified cases, the network focused on these important retinal areas, macula, blood vessel arcs, and lesions-per-case features such as microaneurysms and hemorrhages. Misclassified images would show more dispersed or misplaced activation; for instance, around the optic disc or along vessel branches but without exhibiting clear lesions.

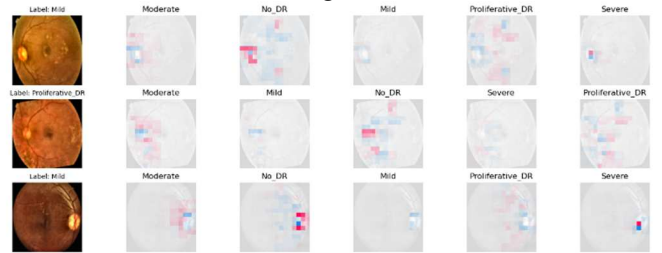


Figure 5 SHAP visualizations for images

Figure 5 shows SHAP-based pixel-wise attributions on selected validation images further confirmed that positive contributions from the model mainly come from dense areas of lesions; background and non-pathological regions either produce negative or weak effects in the predicted grade. These interpretability results provide qualitative evidence of the meaningful retinal structure that primarily supports the model's decisions, rather than trivial artifacts.

V. DISCUSSION

A. Analysis of Overfitting

Relative to earlier runs without dropout and weight decay, the present fine-tuning curves exhibited a reduced training—validation loss gap throughout learning. Validation loss decreased in parallel with training loss rather than flattening or rising mid-training, and validation accuracy and weighted F1 improved monotonically. These observations indicated that dropout ($p = 0.5$) and L2 weight decay ($k = 10^{-4}$) constrained the effective capacity of the head and discouraged over-

specialization to the training distribution, yielding a more generalizable model.

B. Error Analysis

The confusion matrix showed that misclassifications concentrated between adjacent severities. The most common error was under-grading Moderate as Mild, followed by over-grading Moderate toward Proliferative DR; misclassification between No DR and diseased classes was rare. This pattern is consistent with borderline phenotypes in which the burden of microaneurysms, hemorrhages, and exudates lies near grading thresholds. Potential mitigations include ordinal-aware objectives, class-balanced or focal losses, curriculum strategies that emphasize boundary cases, and targeted augmentation mimicking subtle lesion patterns.

C. Limitations

A major limitation was the small sizes of the classes in the original dataset. Due to this small size, certain classes included augmented images in the testing dataset, which may have led to slightly higher validation metrics. Furthermore, evaluation was performed on a single publicly sourced dataset with image-wise splits; the absence of patient identifiers prevented patient-wise partitioning and may permit information leakage across splits. Moreover, external validity across camera types, acquisition protocols, and demographics remains untested. In addition, training relied on class balancing and class-weighted loss, which may distort prevalence and calibration if applied naively in deployment settings. Also, only a single random seed was reported; uncertainty estimates and robustness to initialization were not quantified.

VI. CONCLUSION AND FUTURE WORK

This study addressed automated grading of diabetic retinopathy from color fundus images using a transfer-learning approach with ResNet-50, a principled preprocessing pipeline, dataset balancing, and class-weighted optimization. The resulting model achieved QWK 0.82, accuracy 0.85, and micro-average AUC 0.98 on a stratified validation split, with high sensitivity and specificity for referable DR. These results indicated that the proposed framework produced clinically meaningful discrimination while controlling overfitting. A contribution of this study is the combination of both SHAP and Grad-CAM techniques in analysis. Regarding future work, we plan to (i) evaluate ordinal-aware (e.g., CORAL, cumulative-link) and focal/class-balanced losses to reduce Moderate under-grading; (ii) incorporate multi-seed evaluation and cross-device/cohort external validation; (iii) study calibration and uncertainty to align with screening priorities; (iv) investigate self-supervised/domain-specific pretraining and higher-resolution inputs; and (v) add multi-task lesion detection/segmentation heads to encourage pathology-aware features.

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